Life-Cycle Decision Support for Post-Optimization Industrial Decarbonization: An Environmental-Informatics Case Study of an SME Foundry

Ahmed El-Feky[1]*, Volker Wohlgemuth[2], and Amna Ramzy[1]

[1] Faculty of Engineering and Materials Science, German University in Cairo, Cairo, Egypt

[2] Faculty 2: Engineering – Technology and Life, HTW Berlin – University of Applied Sciences, Berlin, Germany
*Corresponding author: Ahmed El-Feky (ahmed.el-feky@guc.edu.eg)

Abstract

Energy-intensive small and medium-sized enterprises need decarbonization pathways that connect operational improvement with life-cycle evidence for the remaining demand. This paper presents an environmental-informatics workflow for further decarbonization after material and energy flow optimization in an Egyptian SME foundry. The workflow links plant data collection, material and energy flow analysis, hotspot-based optimization, photovoltaic scenario modelling, and life cycle assessment into a sequential decision-support structure. Operational optimization first reduces avoidable electricity demand by 430,312 kWh/year, lowering grid-related climate impacts by about 288 t CO_2-eq/year and reducing electricity consumption by 22%. Life cycle assessment is then applied to the optimized residual electricity demand of approximately 1.90 GWh/year, comparing continued grid supply with a hybrid rooftop photovoltaic–grid scenario. Using a P75 photovoltaic yield of about 1.62 GWh/year, a grid emission factor of 0.67 kg CO_2-eq/kWh, and a photovoltaic life-cycle intensity of 0.04 kg CO_2-eq/kWh, the hybrid photovoltaic–grid scenario avoids approximately 28.5 kt CO_2-eq over the 30-year assessment period compared with the optimized grid-only case. The optimized annual climate impact is reduced from about 1,275 to 252 t CO_2-eq/year. The non-climate indicators show strong reductions in fossil resource scarcity and water depletion, but also an increase in mineral resource scarcity due to the material requirements of photovoltaic components, including modules, inverters, mounting structures, and cabling. These results demonstrate why LCA is required after process optimization: it quantifies the additional decarbonization potential of cleaner electricity supply, reveals residual environmental hotspots, and prevents a climate-only interpretation of renewable-energy integration.

Keywords Life cycle assessment - Environmental informatics - Industrial decarbonization - Material and energy flow analysis - SME foundry - Photovoltaic electricity

Introduction

Industrial decarbonization is a central requirement for limiting climate change because industry remains one of the largest contributors to global energy demand and greenhouse-gas emissions [1, 2]. In fossil-fuel-dominated power systems, electricity-intensive factories face a particularly difficult transition: even when direct combustion inside the facility is limited, purchased electricity can create a high indirect carbon footprint [3]. For this reason, renewable electricity integration, especially rooftop photovoltaic (PV) systems, is often presented as a direct substitution pathway: replace grid electricity with lower-carbon electricity and report the avoided emissions [4]. Although this approach is useful, it is incomplete because it treats industrial decarbonization mainly as an energy-supply problem.

For electricity-intensive manufacturing sites, avoidable demand and residual demand are two different decision problems. Avoidable demand refers to energy that can be reduced through operational improvements, better scheduling, material-flow control, or process optimization. Residual demand refers to the remaining electricity requirement after these efficiency measures have been applied. If renewable-energy scenarios are evaluated before process optimization, the facility may oversize the intervention and miss lower-cost operational measures. Conversely, if optimization is evaluated without life cycle assessment (LCA), the analysis may understate the environmental consequences of the remaining energy system and overlook burden shifting across non-climate indicators such as fossil resource scarcity, mineral resource scarcity, and water depletion [5, 6]. LCA provides a structured method to evaluate these wider environmental consequences because it considers impacts across the full life cycle, from raw material extraction and component manufacturing to operation and end-of-life treatment [7, 8]. This is particularly important for PV-based decarbonization, where operational greenhouse-gas emissions are low, but upstream manufacturing, metals, inverters, mounting systems, cabling, transport, and end-of-life processes can influence resource-related impact categories [9]. Therefore, LCA is not only a tool for confirming climate benefits; it is also a decision-support method for identifying trade-offs and preventing a climate-only interpretation of renewable-energy integration [10]. This issue is especially relevant for small and medium-sized enterprises (SMEs). SMEs represent a major part of industrial activity and can have substantial cumulative environmental impacts, but they often have fewer financial, technical, and human resources for advanced sustainability assessment than larger companies [11, 12]. As a result, SMEs require transparent and practical modelling workflows that can transform available plant data into decision-ready sustainability evidence. In energy-intensive foundries, this need is even more critical because induction melting, machining, compressed-air systems, auxiliary equipment, and production scheduling create a strong dependency on electricity. The decarbonization question should therefore not be limited to whether rooftop PV electricity is cheaper or cleaner than grid electricity. The more useful question is how LCA can guide the next decarbonization step after operational optimization has already reduced avoidable demand. The conference contribution of this paper is an environmental-informatics workflow rather than a standalone photovoltaic feasibility study. The proposed workflow combines two sequential layers. The first layer uses material and energy flow analysis (MEFA) to map the production system, identify hotspots, and reduce avoidable energy and material losses [13]. The second layer applies LCA to the optimized residual demand in order to compare grid-only and hybrid PV–grid electricity supply, quantify further decarbonization potential, and identify environmental trade-offs [14]. This framing fits the EnviroInfo topic of digitalization and sustainability because the focus is the use of environmental modelling tools to convert site-specific industrial data into transparent, reproducible, and decision-ready evidence for SME decarbonization.

Life Cycle Assessment

Operational indicators are essential for identifying and verifying improvements within the factory boundary, but sustainability assessment often requires a broader perspective to avoid burden shifting across supply-chain stages. Life cycle assessment provides a standardized framework for quantifying environmental impacts across the life cycle of a product or service, from upstream supply chains through operation to end of life. ISO 14040 and ISO 14044 define the core phases (goal and scope definition, inventory analysis, impact assessment, and interpretation) and emphasize transparency of assumptions and boundaries to support consistent use and comparability [5,6].

In practice, life cycle assessment translates inventory flows (emissions and resource extractions) into impact indicators using life cycle impact assessment methods. ReCiPe 2016 is widely used because it provides harmonized midpoint and endpoint characterization across multiple impact categories, supporting multi-indicator assessment beyond climate change alone [10]. Background inventories are commonly sourced from databases such as ecoinvent, and the ecoinvent literature highlights that system-model choices (for example cut-off, APOS, or consequential) can influence results, reinforcing the need to specify database version and modelling logic explicitly [15, 16]

Flow-Modelling Tools (UMBERTO®) and the Tool Landscape

Software tools play a practical role in sustainability measurement because they support consistent process-network representation, automated balance checking, structured scenario analysis, and traceable documentation of assumptions and boundaries. In flow-based applications, tools are used to build Material and Energy Flow Analysis or Material Flow Analysis networks and generate visual outputs (including Sankey diagrams), while in life-cycle applications tools are used to construct life cycle models and calculate impact indicators using standardized databases and characterization methods [5,6], [17]. In practice, the tool landscape can be viewed as two complementary categories: (i) flow analysis and resource-efficiency tools for plant-level transparency, hotspot identification, and operational optimization, and (ii) life cycle assessment tools for ISO-aligned cradle-to-grave modelling and multi-indicator impact reporting. Within the UMBERTO® software family, these needs are addressed by distinct products. UMBERTO® Efficiency+ is typically used for operational flow modelling and improvement scenarios inside the factory boundary, supporting auditable baselining and hotspot-driven optimization with Sankey-based visualization. UMBERTO® Life Cycle Assessment is designed for life cycle applications, enabling cradle-to-grave modelling with background datasets (for example ecoinvent) and life cycle impact assessment methods, consistent with ISO 14040 and ISO 14044 reporting expectations. In other words, Efficiency+ is primarily an operational transparency and optimization environment, while Umberto Life Cycle Assessment is the life cycle modelling and impact assessment environment; used together, they support consistent measurement across operational and life-cycle perspectives [5, 6], [17]. Beyond Umberto, widely used life cycle assessment software ecosystems include SimaPro and openLCA, both positioned as professional tools for modelling product systems and calculating impacts using established databases and method [18, 19]. While tools differ in licensing, interface design, and database workflows, the key point for this research is functional: flow tools are strongest for internal process transparency and hotspot-based optimization, while life cycle tools are required for cradle-to-grave accounting and multi-indicator reporting to avoid burden shifting.

Environmental Indicators

The environmental assessment focuses on four indicators that are directly relevant to the study objective and to the expected trade-offs between grid electricity and rooftop photovoltaic electricity. These indicators are climate change, fossil resource scarcity, mineral resource scarcity, and water depletion. Together, they allow the analysis to move beyond a carbon-only comparison and to examine whether further decarbonization creates additional environmental burdens in other impact categories.

Table 1: Environmental indicators considered in this study

Indicator	Definition	Unit
Climate change	Potential contribution to climate change from life-cycle greenhouse gas emis-	kg CO_2-

	sions	eq
Fossil resource scarcity	Potential depletion of fossil resources across fuel extraction, electricity generation, and related supply chains	MJ oil-eq
Mineral resource scarcity	Potential scarcity of mineral resources associated with metals and materials used in system components	kg Cu-eq
Water depletion	Potential freshwater consumption from upstream manufacturing, electricity generation, operation, and maintenance activities	m^3

Climate change is used as the main indicator for evaluating greenhouse-gas mitigation when part of the optimized electricity demand is supplied by rooftop PV instead of the grid. Fossil resource scarcity reflects the dependence of grid electricity on natural gas, oil, and other fossil-based supply chains, and therefore helps quantify the resource benefit of reducing grid imports. Mineral resource scarcity is included because PV systems require material-intensive components, such as modules, inverters, mounting structures, and cabling. This indicator is important for identifying whether climate benefits are accompanied by higher demand for metals such as copper and aluminum. Water depletion is also considered because the case study is located in Egypt, where water availability is a relevant sustainability concern. It captures freshwater use linked to upstream manufacturing, electricity supply, and PV operation, including module cleaning.

Aim and Objectives

Building on the previously completed Phase 1 study, "Optimizing Foundry Operations: A Case Study on Improving Material and Energy Flow Efficiency in Line with the SDGs" [20], this paper focuses on the next decarbonization step. The earlier study used UMBERTO® Efficiency+ to identify operational hotspots in the case-study foundry and reported annual savings of approximately 430,312 kWh of electricity, 288 t CO_2, and 11,100 kg of recovered metallic iron [20]. Therefore, the present paper does not re-present material and energy flow optimization as a new contribution. Instead, it uses the optimized foundry demand as the starting point for evaluating how life cycle assessment can guide further decarbonization of the remaining electricity demand.
The aim of this paper is to demonstrate how life cycle assessment can be used as a post-optimization decision-support layer to assess further decarbonization potential, identify environmental trade-offs, and support cleaner electricity-supply decisions for an energy-intensive SME foundry.

To achieve this aim, the paper pursues the following objectives:

- To use the optimized electricity demand from the previously published Phase 1 study as the reference case for further decarbonization assessment.
- To define and evaluate a post-optimization electricity-supply comparison between an optimized grid-only case and a hybrid photovoltaic–grid case using cradle-to-grave life cycle assessment.
- To compare both electricity-supply cases across climate change, fossil resource scarcity, mineral resource scarcity, and water depletion in order to identify additional decarbonization benefits, residual hotspots, and possible environmental trade-offs beyond operational CO_2 reduction.

Research Question

This study is guided by the following research question: How can life cycle assessment support further decarbonization of an optimized energy-intensive SME foundry in Egypt through rooftop photovoltaic integration, while capturing environmental trade-offs beyond operational CO_2 reduction?

SDG Alignment

The contribution is aligned with SDG 7 on affordable and clean energy, SDG 9 on industry, innovation and infrastructure, SDG 12 on responsible consumption and production, and SDG 13 on climate action, because it links industrial energy optimization, renewable electricity integration, and life-cycle-based environmental decision support for an energy-intensive SME.

Methodology

This paper applies a post-optimization life cycle assessment methodology to evaluate the remaining decarbonization potential of an energy-intensive SME foundry in Egypt. The methodological logic is based on an efficiency-first sequence. The operational optimization stage was already completed and reported in the previous Phase 1 study [20]. Therefore, the present paper starts from the optimized electricity demand and focuses on the environmental consequences of supplying this residual demand through alternative electricity-supply configurations. The overall methodological structure is shown in Fig. 1 and consists of four main steps: defining the optimized reference case, modelling electricity-supply scenarios, applying cradle-to-grave LCA, and interpreting the results across climate and non-climate indicators.

Case-Study Basis and Optimized Reference Demand

The case study is an electricity-intensive SME foundry located in Egypt. The foundry produces ductile iron and steel castings and relies mainly on electricity for induction melting, machining, auxiliary equipment, and supporting production operations. In the previously published Phase 1 study, material and energy flow modelling was used to identify major operational hotspots and to reduce avoidable electricity demand through process optimization [20]. That study reported an annual electricity saving of approximately 430,312 kWh and the recovery of around 11,100 kg of metallic iron through operational improvement measures [20]. In the present paper, these optimization results are not recalculated because the measures were already applied, tested, and reported in the Phase 1 study [20]. Instead, the verified optimized demand is used as the starting condition for the post-optimization assessment. This allows the present study to focus specifically on the next decarbonization step: evaluating how life cycle assessment can guide cleaner electricity-supply decisions for the remaining electricity demand.

Goal and Scope Definition

Two post-optimization electricity-supply scenarios are evaluated. The first is the **optimized grid-only case**, where the residual electricity demand after the Phase 1 optimization is supplied entirely by the Egyptian medium-voltage grid. This scenario represents the improved operational condition of the foundry before introducing any supply-side decarbonization measure. The second is the **hybrid photovoltaic–grid case**, where a 1,000 kWp rooftop PV system supplies a substantial share of the optimized residual demand, while the remaining electricity is imported from the grid. The PV system is modelled under net-metering conditions, in which surplus generation can be exported to the grid and later credited against electricity imports. This enables the rooftop PV system to be assessed as part of the foundry's electricity-supply mix rather than as a standalone generation unit. This scenario design isolates the environmental effect of renewable electricity integration after operational optimization. Unlike a conventional PV feasibility study, the rooftop PV system is not assessed against the original unoptimized baseline. Instead, it is evaluated against the verified optimized residual demand, allowing the analysis to quantify the additional life-cycle decarbonization potential achieved after efficiency improvements have already been implemented.

Functional Unit

The functional unit (FU) is defined as 1 kWh of electricity delivered to the foundry at the point of common coupling (PCC) over the 30-year assessment period. In practice, the PCC corresponds to the bi-directional net-metering (utility) meter, which records both electricity imported from and exported to the distribution grid. Defining the FU at this interface ensures a consistent comparison of environmental intensities (e.g., kg CO_2-eq/kWh) between the grid-only and Hybrid scenarios.

Scenario Definition and Net Metering

Three PV yield scenarios, P50, P75, and P90, were considered to account for interannual variability and uncertainty in annual electricity generation. The P75 case, with an expected annual generation of approximately 1.62 GWh/year, was selected as the central scenario because it provides a moderately conservative but realistic basis for environmental interpretation. A 30-year operating lifetime and an annual degradation rate of 0.5% were used for the main life-cycle assessment. Table 2 summarizes the role of each PV yield scenario in the study.

Table 2: Annual PV generation scenarios (P50, P75, P90): interpretation and use in this study

Scenario	Description	Role in Study
P50	Median expected annual generation	Most likely (benchmark) estimate
P75	Moderately conservative annual generation	***Central/base-case scenario***
P90	Highly conservative, low-yield outcome	Downside / robustness check

Selecting **P75** as the reference scenario ensures that environmental results are reported under **cautiously conservative but realistic** operating conditions, under Egypt's net-metering framework.

Life Cycle Inventory

The life cycle inventory combines site-specific electricity demand data with background datasets for electricity generation and PV system components shown in Table 3. The optimized annual demand is derived from the Phase 1 optimization results [20]. PV electricity generation is based on the rooftop PV design and annual yield assumptions used for the case-study facility. The PV system includes modules, inverters, mounting structures, cabling, cleaning water, maintenance activities, and end-of-life treatment. The life cycle model is implemented using Umberto Life Cycle Assessment with ecoinvent 3.11 background data. The ecoinvent database is used because it provides structured and traceable inventory datasets for electricity supply, PV components, material production, transport, and end-of-life processes. The database version and system model are reported explicitly to support transparency and reproducibility, since inventory modelling choices can influence LCA results [15], [16].

Table 3: Life cycle inventory mapping for the 1,000 kWp rooftop PV system

No.	Life-cycle stage	Component / activity	ecoinvent v3.11 dataset	Unit	Quantity
1	Component pro-	Photovoltaic modules	market for photovoltaic panel, single-	m²	4,350

	duction		Si wafer – GLO		
2	Component production	AC/DC cabling	market for cable, three-conductor cable – GLO	m	14,000
3	Component production	Aluminum mounting structure	market for section bar extrusion, aluminum – GLO	kg	50,000
4	Component production	Inverters, including replacement	market for inverter, 500 kW – GLO	unit	3.5
5	Installation	Roof-mounted PV installation	installation, photovoltaic plant, roof-mounted – GLO	kWp	1,000
6	Operation and maintenance	Cleaning water over 30 years	tap water – RoW proxy	m^3	1,080
7	Operation and maintenance	Maintenance travel over 30 years	transport, passenger car – GLO	km	4,500
8	End of life	Silicon-based PV module recycling	treatment of waste photovoltaic panel, silicon-based, recycling – GLO	kg	51,000
9	End of life	Aluminum mounting-structure recycling	treatment of aluminum scrap, recycling – GLO	kg	50,000

PV Performance Modelling in PVsyst (v8)

The proposed 1 MWp grid-connected rooftop PV system was modelled in PVsyst v8.0.18 for 10th of Ramadan, Egypt (30.22°N, 31.78°E; 161 m; UTC+2). Meteorological inputs were taken from Meteonorm v8.2 (1996–2015) using a synthetic dataset (satellite fraction 46%). near-shading losses were not modelled, due to the absence of significant rooftop obstructions. The PV array comprises 1,600 modules rated at 625 Wp (TWMND-78HD-625), producing 1,000 kWp DC, configured as 100 strings × 16 modules. The inverter stage uses seven SG125CX-P2 units for a total of 875 kW_AC, corresponding to a DC/AC ratio of 1.14. The system is modelled as a fixed-tilt plane with 20° tilt and 0° azimuth (south-facing). PVsyst predicts *1,622,871 kWh/year* ***(****1,623 kWh/kWp/year****,*** *79.76%* performance ratio), and this output is used consistently in the life cycle models [21], [22].

Life Cycle Impact Assessment

Life cycle impact assessment is conducted using the ReCiPe 2016 Midpoint method. Four indicators are selected because they are directly relevant to the comparison between grid electricity and PV-based electricity supply: climate change, fossil resource scarcity, mineral resource scarcity, and water depletion. Climate change is used to quantify greenhouse-gas mitigation. Fossil resource scarcity captures the reduction in fossil fuel dependence when grid electricity is partly displaced. Mineral resource scarcity captures material-related trade-offs from PV modules, inverters, mounting structures, and cabling. Water depletion is included because the case study is located in Egypt, where water availability is an important sustainability concern. The impact results are reported per functional unit and then scaled to annual and 30-year values for interpretation. The 30-year assessment period reflects the assumed operating lifetime of the rooftop PV system. This allows the study to report both the annual reduction in climate impact and the cumulative avoided emissions over the full assessment period.

Implementation as a reproducible information workflow

The workflow was structured as a set of information objects rather than as a single calculation sheet. This is important for reproducibility because each modelling layer has a different data role. Plant invoices and metered electricity data define the annual electricity demand. Process-flow information defines where electricity is used and where optimization is feasible. PVsyst output defines the expected renewable generation under P50, P75 and P90 production conditions. LCA inventory mapping defines the supply-chain burdens of grid electricity and PV components. Finally, interpretation tables translate numerical results into decision consequences for managers.

Treating these objects separately reduces the risk of mixing physical performance, and environmental impacts. For example, PV yield affects the quantity of grid electricity displaced, but it does not change the manufacturing burden per unit of installed PV capacity. The proposed information architecture keeps these assumptions traceable and allows the same case to be updated when better plant data, a newer ecoinvent version or a different grid mix becomes available.

For an SME user, the workflow can be implemented without building a complex enterprise platform. A spreadsheet can hold the cleaned plant data and annual scenarios, MEFA software can represent the production system and hotspot structure, PVsyst can generate annual yield scenarios, and Umberto LCA or another ISO-compliant LCA environment can calculate midpoint impacts. The added value is not the brand of software, but the disciplined handover between tools: demand after optimization becomes the input to the energy-supply LCA, and LCA interpretation becomes the input to the final decarbonization decision.

Table 4: Information objects used to make the MEFA-LCA workflow transparent and updateable

Information object	Main input	Model role	Decision output
Plant operation data	Invoices, meters, production records	Defines baseline and optimized demand	Annual energy saving and residual load
MEFA model	Process map and material-energy flows	Identifies avoidable losses and hotspots	Optimization actions before PV sizing
PV scenario model	Roof area, system capacity, P75 yield	Estimates renewable supply potential	PV-grid energy balance
LCA model	Grid dataset, PV inventory, LCIA method	Quantifies life-cycle impacts	Climate benefit and resource trade-offs
Interpretation layer	Indicator changes and assumptions	Links numbers to decisions	Efficiency-first decarbonization pathway

Data Sources, Use, and Validation

The assessment combines previously verified plant-optimization results with site-specific PV design data and background life cycle inventory datasets. The optimized electricity demand is taken from the completed Phase 1 study, where the operational improvement measures were applied, tested, and reported [20]. Therefore, the present paper does not rebuild the original material and energy flow baseline. Instead, it uses the verified optimized residual demand as the reference condition for comparing the optimized grid-only and hybrid PV–grid electricity-supply cases. Site-specific data were used wherever available, including the optimized electricity demand, rooftop PV capacity, main system components, and PV generation assumptions. Where direct plant data were unavailable, literature-based and database-supported assumptions were used, including PV lifetime, degradation, inverter replacement, background electricity supply, component inventories, and life cycle impact characterization. Table 5 summarizes the provenance and use of the main data inputs. This distinction between verified plant data, technical design assumptions, simulation outputs, and background LCA datasets supports transparency and reproducibility.

Table 5: Sources and uses of the main data inputs

Data item	Source type	Used for
Optimized annual electricity demand	Previously completed Phase 1 study [20]	Reference demand for the post-optimization LCA
Annual electricity saving after optimization	Verified Phase 1 optimization result [20]	Distinguishing avoidable demand from residual demand
Optimized grid-only electricity supply	Calculated from optimized residual demand and Egyptian grid dataset	Reference electricity-supply scenario
Rooftop PV capacity	Site-specific system design / technical specification	Definition of the hybrid PV–grid scenario
PV modules, inverters, cabling, and mounting structure	Supplier specifications and technical assumptions	Foreground inventory for PV system components
PV annual yield and performance assumptions	PVsyst-based simulation assumptions	Annual PV electricity generation in the hybrid scenario
PV degradation, lifetime, and inverter replacement	Literature and technical assumptions	30-year LCA modelling of PV electricity supply
Cleaning water and maintenance travel	Operational assumptions for the PV system	Operation and maintenance inventory
End-of-life treatment of PV modules and metals	ecoinvent v3.11 background datasets	End-of-life modelling and recycling-related impacts
Egyptian medium-voltage grid electricity	ecoinvent v3.11 background dataset	Optimized grid-only case and residual grid imports
PV component and material datasets	ecoinvent v3.11 background datasets	Life cycle inventory of the rooftop PV system
ReCiPe 2016 Midpoint (H) indicators	Life cycle impact assessment method	Climate change, fossil resource scarcity, mineral resource scarcity, and water depletion
30-year assessment period	Actual PV market quotations, supplier technical offers, and warranty assumptions	Annual and cumulative impact interpretation

3. Results and Interpretation

This section presents the outcomes of the rooftop PV integration assessment by comparing two electricity-supply configurations: a grid-only baseline and a rooftop PV–grid hybrid system operated under a net-metering arrangement. The analysis was conducted over a 30-year project lifetime using the P75 PV yield estimate, an annual PV degradation rate of 0.5%. Environmental performance is reported per 1 kWh of electricity supplied to the foundry at the point of common coupling (PCC). Where relevant, selected indicators are also aggregated at annual and lifetime scales to provide a clearer view of the cumulative environmental implications of PV deployment.

Environmental Results: Life Cycle Assessment

The environmental performance of the grid-only and hybrid electricity-supply scenarios was evaluated using a comparative life cycle assessment model developed in Umberto LCA. The analysis employed the ReCiPe 2016 Midpoint (H) impact assessment method together with ecoinvent v3.11 background datasets. The comparison focuses on the impact categories most directly associated with electricity-supply decarbonization and resource use, namely climate change, fossil resource scarcity, mineral resource scarcity, and water depletion. These categories were selected because they capture both the expected benefits of reducing grid electricity consumption and the potential upstream burdens associated with PV module production, balance-of-system components, and system replacement over the project lifetime. The results are first used to quantify the environmental differences between the two electricity-supply configurations. A subsequent contribution analysis is then applied to identify the dominant processes, life-cycle stages, and environmental hotspots within the modeled system. This structure allows the assessment to distinguish between overall impact reduction at the PCC and the specific sources of residual environmental burden in the rooftop PV–grid hybrid scenario.

LCA Contribution Analysis and Model Outputs

Figure 1 below presents the Umberto LCA process network, which serves as the core implementation tool for the PV life-cycle model. Umberto is used to (i) map the site-specific PV bill of materials and activities to the correct ecoinvent 3.11 (cut-off) datasets, creating a traceable cradle-to-grave inventory for modules, inverters, mounting, cabling, transport, operation, and end-of-life treatment; (ii) calculate the selected ReCiPe 2016 Midpoint (H) impact indicators for the PV system and then combine PV electricity with the Egyptian medium-voltage grid dataset to quantify the hybrid approach per kWh at the point of common coupling; and (iii) document and verify the model through the process-network view, which makes boundaries, assumptions, and flow linkages explicit. The grid-only baseline is represented directly by the ecoinvent grid electricity dataset, where all impact indicators are already reported per kWh (for example, 0.67 kg CO_2-eq/kWh), ensuring a consistent, comparable functional unit without building a separate grid network model.

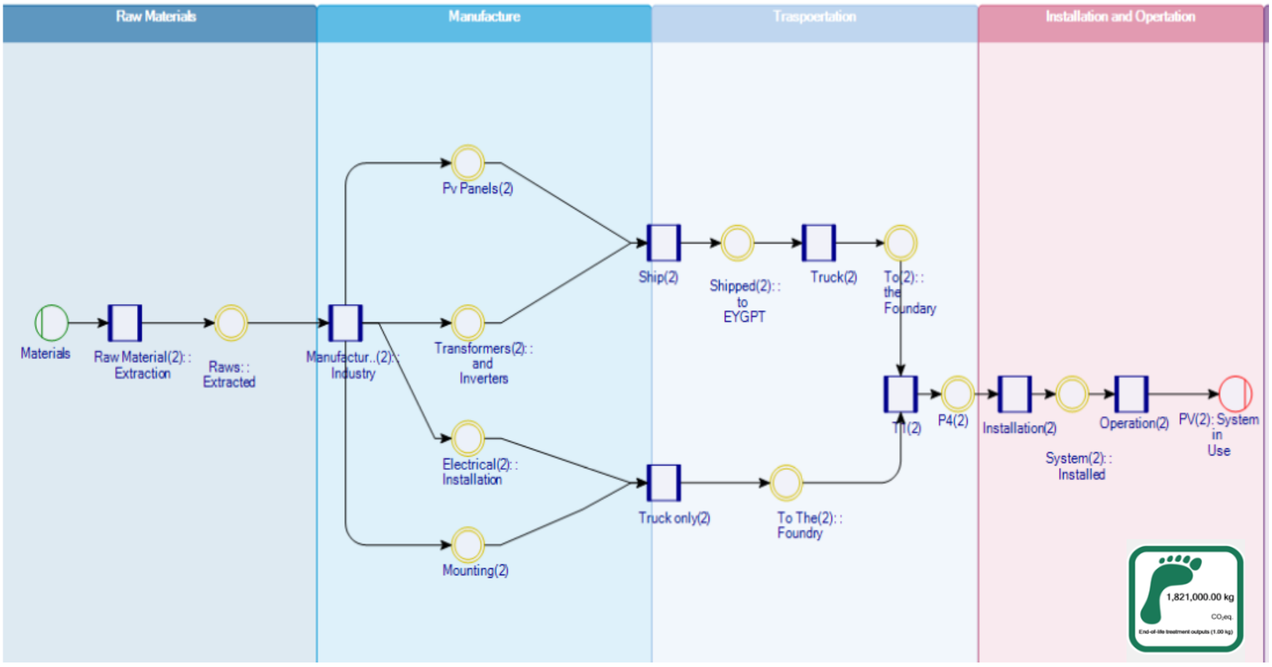

Figure 1: Cradle to-grave life-cycle system boundary of the rooftop PV system

Climate change

The climate-change results indicate a substantial advantage of the hybrid PV–grid configuration over the optimized grid-only case. Supplying the foundry's optimized annual electricity demand of approximately 1.90 GWh entirely from the Egyptian medium-voltage grid results in a climate-change impact of about 1,275 t CO_2-eq/year. In the hybrid case, rooftop PV supplies approximately 1.62 GWh/year, while the remaining demand is supplied by the grid. This reduces the annual climate-change impact to about 252 t CO_2-eq/year, corresponding to an approximately 80% reduction in the first year. Over the 30-year assessment period, the hybrid PV–grid scenario avoids approximately 28.5 kt CO_2-eq compared with the optimized grid-only case. Residual grid electricity remains the dominant contributor to climate impact in the hybrid scenario, while upstream PV manufacturing contributes a smaller but visible share. Table 6 reports the annual and cumulative climate-change results, and Figure 2 visualizes the cumulative reduction over the 30-year assessment period.

Table 6: Comparison of annual GWP and emission intensity for grid-only versus Hybrid supply scenarios

Scenario	PV Energy (kWh/year)	Grid Energy (kWh/year)	GWP Factors (kg CO_2-eq/kWh)	Total GWP (kg CO_2-eq/year)	GWP Intensity (kg CO_2-eq/kWh)	Cumulative GWP over 30 years (kg CO_2-eq)	Change vs Grid-Only
Grid-Only Baseline	0	1,902,833	Grid = 0.67	1,273,000	0.670	38,190,000	—
Hybrid — Year 1	1,622,871	279,173	PV = 0.04; Grid = 0.67	251,879	0.133	—	80.2% lower in Year 1

Hybrid — 30-year average	1,507,879	392,121	PV = 0.04; Grid = 0.67	323,036	0.170	9,691,086	74.6% lower over 30 years

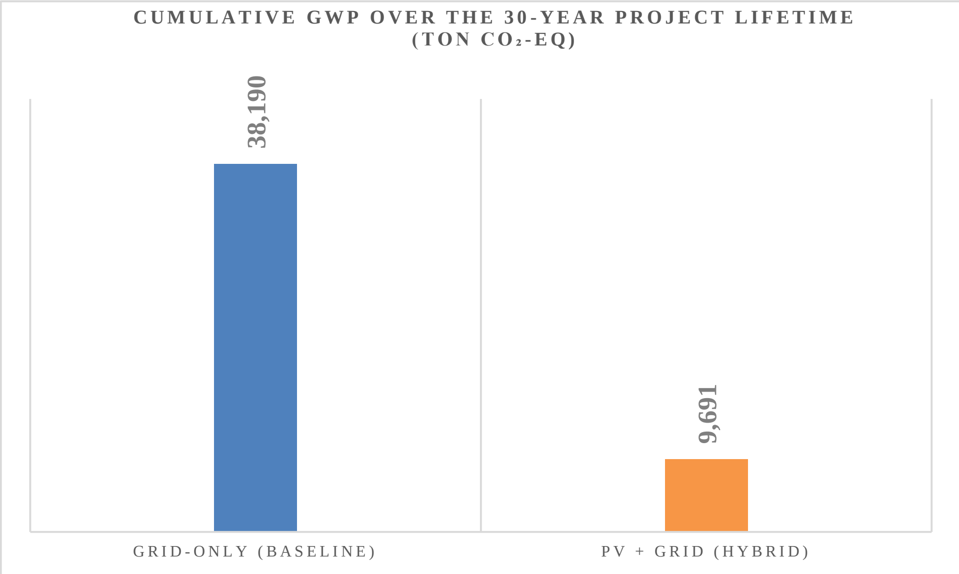

Figure 2: Thirty-year cumulative climate-change impact of the optimized grid-only and hybrid PV–grid scenarios.

Fossil Resource Scarcity, Mineral Resource Scarcity, and Water Depletion

At the system level, the hybrid PV–grid scenario substantially lowers fossil resource use and water depletion relative to the optimized grid-only case, while introducing a clear burden shift toward mineral resource demand. This pattern is consistent with photovoltaic LCA literature, where climate and fossil-resource benefits are often accompanied by higher material-related impacts. The trade-off can be reduced through durable components, efficient system design, responsible material procurement, and credible recycling routes for PV modules, inverters, mounting structures, and cabling. Overall, the results confirm that rooftop PV should not be evaluated only through avoided operational emissions; it should be assessed through a multi-indicator life-cycle perspective.

Table 7: Relative change in non-climate impacts with PV vs. grid electricity

Impact Indicator	Unit	Grid Electricity (Medium Voltage, Egypt)	PV System (Rooftop, the SME foundry)	Change with PV
Fossil Resource Scarcity	MJ oil-eq	10.3	0.8	↓ 92%
Water Depletion	m^3	0.0013	0.0003	↓ 77%

Mineral Resource Scarcity	kg Cu-eq	0.0001	0.0025	↑ 2400%

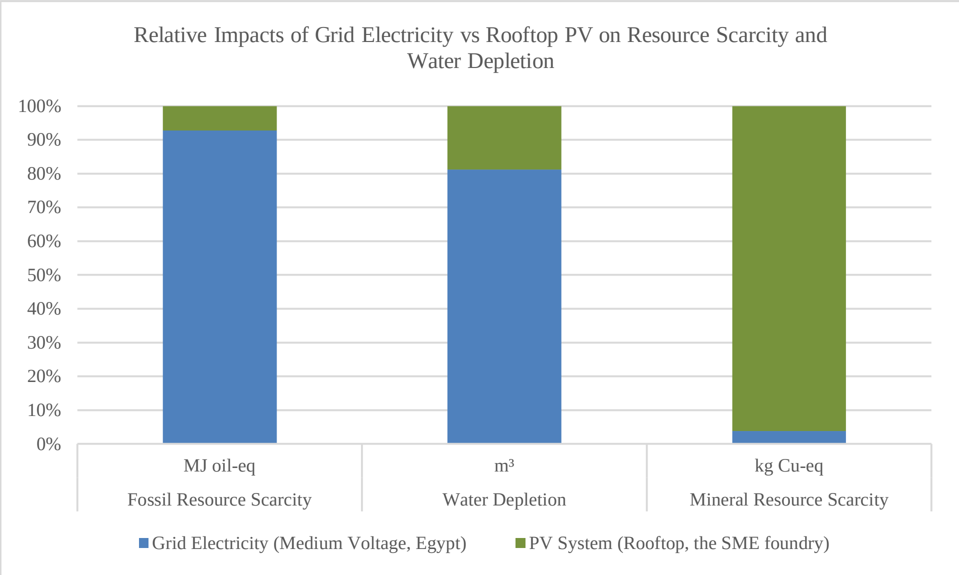

Figure 3: Relative change in selected non-climate indicators for rooftop PV electricity compared with Egyptian medium-voltage grid electricity.

Conclusion

This paper examined how life cycle assessment can support further decarbonization of an optimized energy-intensive SME foundry in Egypt through rooftop photovoltaic integration. Building on the previously completed Phase 1 optimization study, the analysis did not return to the original unoptimized baseline. Instead, the verified optimized electricity demand was used as the reference condition for evaluating the next decarbonization step. This distinction is essential because, after optimization, the foundry no longer represents an inefficient baseline; it represents an improved system with a residual electricity demand that still requires environmental assessment and further decarbonization. The results show that the main value of applying LCA after optimization is not only the calculated CO_2 reduction, but also the change in decision logic. Operational optimization first reduces avoidable demand, while LCA then evaluates the environmental consequences of supplying the remaining demand. This avoids overstating the role of photovoltaic integration by applying it to electricity demand that should first be reduced through operational measures. In the optimized grid-only case, the residual electricity requirement remains a major climate hotspot. In the hybrid PV–grid case, rooftop PV provides substantial additional decarbonization, avoiding approximately 28.5 kt CO_2-eq over the 30-year assessment period and reducing the optimized annual climate impact from about 1,275 to 252 t CO_2-eq/year. The multi-indicator results confirm that renewable electricity integration should not be interpreted only through avoided operational emissions. The hybrid PV–grid scenario reduces fossil resource scarcity and water depletion by lowering dependence on fossil-based grid electricity. However, the increase in mineral resource scarcity shows that PV integration also creates material-related trade-offs linked to modules, inverters, mounting structures, and cabling. This demonstrates why LCA is required as a post-optimization decision-support layer: it quantifies further decarbonization potential, identifies residual hotspots, and prevents a narrow climate-only interpretation of the selected intervention. The study therefore supports an efficiency-first hierarchy for industrial decarbonization. The first step is to reduce avoidable demand through operational and resource-efficiency measures. The second step is to apply LCA to the optimized residual demand and compare feasible electricity-supply scenarios. The third step is to use the LCA interpretation to refine the intervention, for example by prioritizing durable PV modules, efficient inverters, low-loss electrical integration, responsible material procurement, and credible recycling routes for metals and PV modules. In this way, LCA becomes a design and decision-support tool, not only a reporting method. The main contribution of this paper is a staged environmental-informatics workflow that links verified process optimization with life-cycle-based electricity-supply assessment. The focus is how digital environmental modelling can support further decarbonization after operational improvements have already been implemented and tested. This is particularly relevant for SMEs, where sustainability decisions must often be made under constraints related to cost, data availability, and internal analytical capacity. Although the numerical results are case-specific, the workflow is transferable to other electricity-intensive SMEs with sufficient metered electricity data, process-flow information, and rooftop PV potential. The exact magnitude of avoided emissions and non-climate trade-offs will depend on the local electricity mix, PV performance, degradation assumptions, internal electrical losses, net-metering rules, component inventories, and end-of-life assumptions. Future work should extend the workflow by including hourly load matching between PV generation and industrial demand, uncertainty analysis for grid and PV inventory datasets, and circularity scenarios for PV modules, inverters, mounting structures, and cable recovery.

References

[1] IPCC: Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Intergovernmental Panel on Climate Change, Geneva (2023).

[2] Lamb, W. F., Wiedmann, T., Pongratz, J., Andrew, R., Crippa, M., Olivier, J. G. J., et al.: A review of trends and drivers of greenhouse gas emissions by sector from 1990 to 2018. Environmental Research Letters 16(7), 073005 (2021).

[3] International Energy Agency: Energy Policies Beyond IEA Countries: Egypt 2021. IEA, Paris (2021).

[4] International Renewable Energy Agency: Renewable Energy Outlook: Egypt. IRENA, Abu Dhabi (2018).

[5] International Organization for Standardization: ISO 14040: Environmental Management — Life Cycle Assessment — Principles and Framework. ISO, Geneva (2006).

[6] International Organization for Standardization: ISO 14044: Environmental Management — Life Cycle Assessment — Requirements and Guidelines. ISO, Geneva (2006).

[7] Guinée, J. B. (ed.): Handbook on Life Cycle Assessment: Operational Guide to the ISO Standards. Kluwer Academic Publishers, Dordrecht (2002).

[8] Hellweg, S., and Milà i Canals, L.: Emerging approaches, challenges and opportunities in life cycle assessment. Science 344(6188), 1109–1113 (2014).

[9] Frischknecht, R., Stolz, P., Krebs, L., de Wild-Scholten, M., Sinha, P., Fthenakis, V., Kim, H. C., Raugei, M., and Stucki, M.: Life Cycle Inventories and Life Cycle Assessments of Photovoltaic Systems. IEA PVPS Task 12, International Energy Agency Photovoltaic Power Systems Programme (2020).

[10] Huijbregts, M. A. J., Steinmann, Z. J. N., Elshout, P. M. F., Stam, G., Verones, F., Vieira, M. D. M., Zijp, M., Hollander, A., and van Zelm, R.: ReCiPe2016: A harmonised life cycle impact assessment method at midpoint and endpoint level. International Journal of Life Cycle Assessment 22, 138–147 (2017).

[11] OECD: OECD SME and Entrepreneurship Outlook 2019. OECD Publishing, Paris (2019).

[12] Fawcett, T., and Hampton, S.: Why and how energy efficiency policy should address SMEs. Energy Policy 140, 111337 (2020).

[13] Brunner, P. H., and Rechberger, H.: Handbook of Material Flow Analysis: For Environmental, Resource, and Waste Engineers. 2nd edn. CRC Press, Boca Raton (2016).

[14] Muteri, V., Cellura, M., Curto, D., Franzitta, V., Longo, S., Mistretta, M., and Parisi, M. L.: Review on life cycle assessment of solar photovoltaic panels. Energies 13(1), 252 (2020).

[15] Wernet, G., Bauer, C., Steubing, B., Reinhard, J., Moreno-Ruiz, E., and Weidema, B.: The ecoinvent database version 3, part I: overview and methodology. International Journal of Life Cycle Assessment 21, 1218–1230 (2016).

[16] ecoinvent Association: ecoinvent Database v3.11. ecoinvent Association, Zurich (2024).

[17] iPoint-systems GmbH: Umberto Software for Material Flow Analysis and Life Cycle Assessment. iPoint-systems GmbH, Reutlingen. Accessed 8 July 2026.

[18] PRé Sustainability: SimaPro Life Cycle Assessment Software. PRé Sustainability, Amersfoort. Accessed 8 July 2026.

[19] GreenDelta GmbH: openLCA Life Cycle Assessment Software. GreenDelta GmbH, Berlin. Accessed 8 July 2026.

[20] El-Feky, A., Elshiekh, H., and Ramzy, A.: Optimizing Foundry Operations: A Case Study on Improving Material and Energy Flow Efficiency in Line with the SDGs. EnviroInfo Conference Paper (2024).

[21] PVsyst SA: PVsyst Photovoltaic System Simulation Software, Version 8. PVsyst SA, Satigny, Switzerland. Accessed 8 July 2026.

[22] Meteotest AG: Meteonorm Version 8.2: Global Meteorological Database for Solar Energy Applications. Meteotest AG, Bern, Switzerland. Accessed 8 July 2026.